Dianna E. Ochoa Lynch

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https://dianocly.github.io/Dianna-Ochoa-Lynch-portfolio/

Education

Massachusetts Institute of Technology (MIT) | Cambridge, MA *Expected May 2028*
Candidate for B.S. in Mechanical Engineering & Minor in Archeology and Materials

- ***GPA** 4.9/5.0*
- **Relevant Coursework:** Mechanics and Materials II, Thermofluids Engineering II, Design and Manufacturing II, Dynamics and Controls II, Numerical Computation

Technical Skills

- **Mechanical Design:** Precision Mechanism Design, Tendon-Driven Actuation, Weight Optimization, Gearbox Design, Design for Assembly/Manufacturing (DFA/DFM)
- **Precision Manufacturing:** GD&T (Tolerance Selection & Fits), CNC (Manual/CAM), Precision Metrology, 3D Printing, Manual Machining (Mill/Lathe), Laser Cutting
- **Analysis & Simulation:** Finite Element Analysis, MATLAB, Kinematic Modeling, Root Cause and Failure Analysis
- **Software:** CATIA, SolidWorks, Onshape, Inventor Professional, Fusion 360, Arena, Fulcrum, Salesforce

Experience

Mechanical Engineering Intern - Boston Dynamics Waltham, MA
Atlas Actuation *Jun 2026–Aug 2026*

- Redesigned injection-molded actuator components and collaborated with a supplier on sourcing, reducing actuator plastic-part costs by 60%
- Led material selection for actuator-internal plastic components, balancing operating temperature range, chemical compatibility, and injection-molding requirements
- Investigated actuator failures during assembly through root cause analysis and failure-data tracking; developed diagnostic guides and documented findings to support design improvements.
- Co-designed a soldering fixture with a senior engineer to support manufacturing R&D for actuator technology
- Designed and fabricated an actuator cross-sectional model for internal demonstrations and technical instruction

Robotics Researcher - Distributed Robotics Laboratory, CSAIL, MIT Cambridge, MA
Team: *Reporting to Prof. Daniela Rus and PhD Student Wil Norton* *Sep 2024–May 2026*
Project: *Tele-Hand Mk 1* — Anthropomorphic tendon-driven robotic hand for teleoperation and haptic feedback

- Co-designed multi-DoF capstan tendon transmission to minimize backlash and maximize force fidelity for haptic rendering
- Architecting and prototyping a 126:1 multi-stage capstan gearbox (design intent: ~1000°/s finger rotation); iterating helix geometry, reduction staging, and cable tracking stability
- Designed finger skeletal structure and kinematics for rapid iteration, integrating tendon routing and assembly constraints
- Validated tendon routing and stack-up with 3D-printed prototypes; resolved mechanical interferences before metal fabrication
- Led prototype manufacturing from CAD, drawings, and BOMs through tooling, machining, and assembly, targeting ±0.001 in on critical features
- Evaluating skeletal and transmission design for anthropomorphism (ROM coverage, speed, and 20 N grip-force requirement) to guide design tradeoffs and test plans

Makerspace Lead & STEM Instructor Mollerussa, Spain
MIT Global Teaching Labs *Jan 2026–Feb 2026*

- Led makerspace instruction in rapid prototyping and iterative mechanical design, guiding students from sketches and CAD to physical prototypes.
- Maintained fabrication equipment and supported student use of 3D printers, hand tools, and basic machinery.

Mechatronics Engineer – Lead Designer and Systems Engineer Bloomington, IN
FIRST Robotics Competition Team 3494 & FIRST Tech Challenge Team 11329 *Aug 2021–May 2024*

- Directed the end-to-end design of 6 high-performance robots; ranked Top 10 twice out of ~8000 teams worldwide
- Recognized by General Motors for excellence in form-factor optimization, robustness, and serviceability—directly translating complex mechanical subsystems into a "production-ready" design
- Responsible for robot design under strict constraints and managing their fabrication, assembly, and programming

Awards & Activities

MIT Design and Manufacturing I - Brabham BT46 "Fan Car" Prize – For a well designed and well executed robot in MIT's 2.007
Industrial Design Award (sponsored by General Motors) – 2024 FIRST Robotics Competition World Championship

- Recognized as 1 of 8 winners globally for excellence in form-factor optimization, aesthetics, robustness, and serviceability

2023 FRC Dean's List Finalist: Selected as 1 of 2 students from Indiana to represent the state at the FIRST World Championships

Activities: MIT Women's Ultimate Frisbee, Society of Women Engineers, Makerspace Mentor
Interests: Precision Mechanism Design, Manufacturing, Mechatronics, Soft Robotics and Compliant Mechanisms, Stained Glass